Serverless Computing

Reducing Infrastructure Management in the Cloud



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What Is Serverless?

Definition

A cloud model where the provider runs and scales the servers, and you only write and deploy code.

FaaS (Function-as-a-Service) is the compute part: small functions that run in response to events. Examples: AWS Lambda, Azure Functions, Google Cloud Functions.



No servers to manage

Provisioning, patching and scaling are handled by the cloud provider.



Event-driven

Code runs only when triggered: an HTTP call, file upload, message or timer.



Pay per execution

Billed for the time your code actually runs, not for idle servers.

How FaaS Works



1. Write

Write a small, single-purpose function.



2. Deploy

Upload it and attach a trigger (HTTP, upload, timer).



3. Invoke

An event starts the function in a fresh container.



4. Scale

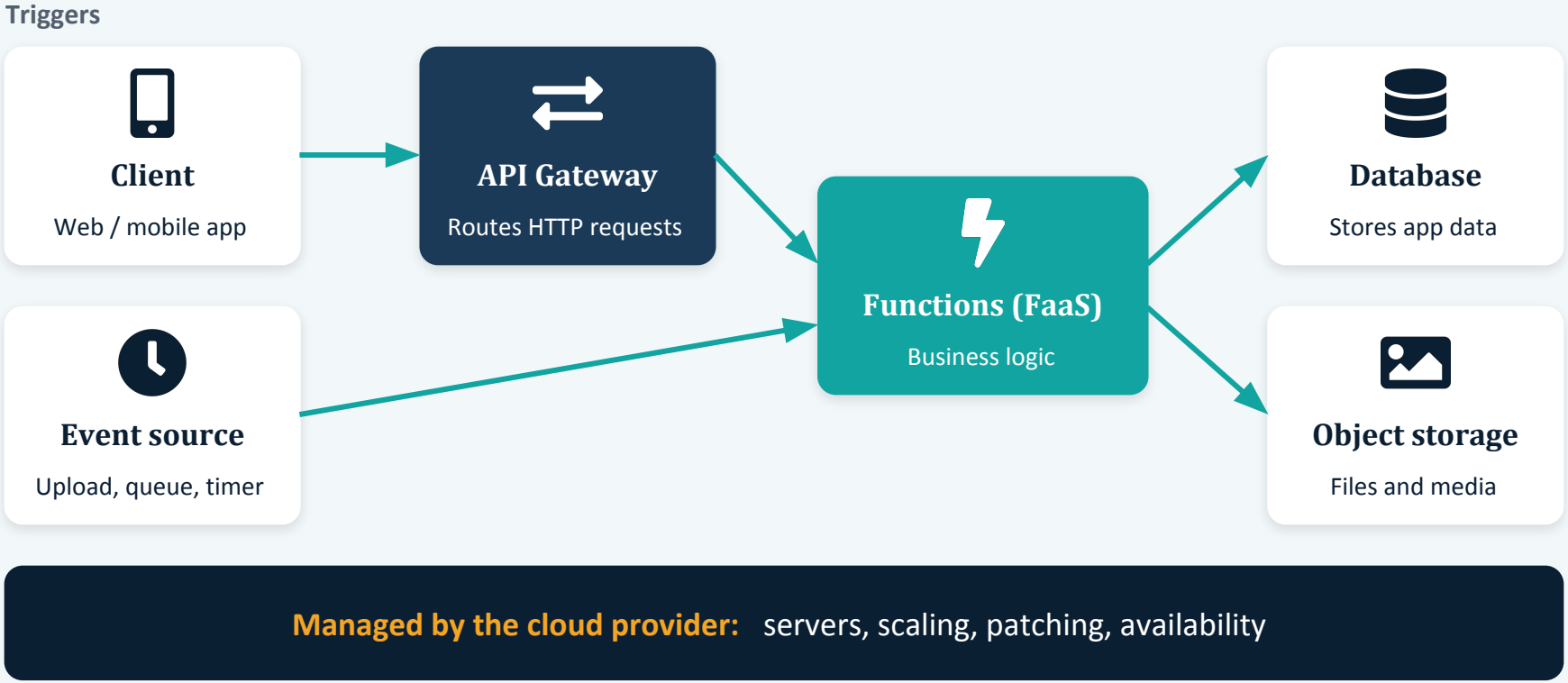
Many events run in parallel; idle means zero.

Stateless
Each run is independent; data lives in a database or storage.

Short-lived
Functions run for seconds or minutes, then stop.

Auto-scaling
Platform adds or removes instances for you.

Serverless Architecture



Managed by the cloud provider: servers, scaling, patching, availability

Business Advantages



Lower cost

Pay only for execution time. No cost for idle servers.



Automatic scaling

Handles traffic spikes without capacity planning.



Faster delivery

Teams ship features sooner by skipping server setup.



Less operations work

No patching or maintenance, so developers focus on the product.

Trade-offs to consider: cold-start delay, vendor lock-in, execution time limits.

Real-World Use Case

Image processing for an online store

1. Seller uploads a product photo
2. Upload event triggers a function
3. Function resizes it into thumbnails
4. Thumbnails saved, record written to database

Why serverless fits

- Runs only when photos arrive
- Scales for sale-day upload spikes
- No cost when nothing is uploaded

Also common: REST APIs, chatbots, IoT data, scheduled jobs.

Conclusion & Key Takeaway



Serverless removes server management; the cloud provider handles it.



FaaS runs small functions on events and scales automatically.



You pay per execution, so cost follows actual usage.

KEY TAKEAWAY

Write code, not servers: serverless lets teams build faster and pay only for what they use.